



Department of Computer Science and Engineering

Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. ECE 'A' Section

Course Code & Title: CS3353 & C Programming and Data Structures

Name of the Faculty member (s): Dr. V. Anusuya, ASCP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit IV / Double Hashing
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO11
- **Activity Chosen:** Jigsaw
- **Justification:**

Double hashing is an important concept in data structures. The different sets of problem given to different groups. The main aim of this Jigsaw activity is to improve the individual and peer learning which requires the students to understand the concept very clearly and engage through the collaborative learning.

- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**

The topic double hashing discussed in the class

- The class divided into 8 teams and each team with 8 members based on the academic performance and ability. The Group named as home group.
- One student from each group selected as a leader. Expert Groups were formed that consist of student from home group.
- Each student in a group is assigned with any one topic and 10 minutes allotted for discussion.

The course instructor delivered the instructions how to perform this activity.

- For collaborative teamwork, the expert group is collapsed and all the students are re-grouped into their home group.
- Now each member in the team will be an expert in any one of hashing resolution techniques. They shared their ideas to all the other members in the group thus enabling each student to learn all the concept.
- This enabled the students to get insight knowledge in this topic.
- After that, the students were asked to discuss the answer for the questions as shown in the figure 1.

• **Images / Screenshot of the practice:**



Figure 1: Students discussion with their team members

• **CO – PO / PSO mapping: PO / PSO mapped:**

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO12	PSO1
CO4	3	3	2	2	1	1	2	1	2

Innovative practice	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO12	PSO1
	3	3	2	2	1	1	2	1	2
Justification for correlation	Requires basic engineering knowledge to implement hash table	Able to Identify engineering problems with hashing resolution techniques	Able to design solutions for complex engineering problems using hashing resolution technique	Able to provide complex problem solutions with hashing techniques	Usage of Visualgo.net tool for performing hashing	Able to solve collision in hashing individually or in team and implement hash table	Able to explain the way to resolve collision in hashing with their team members.	The students can upgrade the technical knowledge using the concept of hashing	Able to apply hash data structures to solve various application in Information & communication Industry

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- All the students were actively participated and done the task and practice among their group members.
 - Each student explored the knowledge of the hashing and hashing resolution concepts among the other students in the class.
 - It helps them to improve the communication skills i.e interacting effectively with the group and summarizing the concept.

- ❖ ***Benefit of the practice:***

- Students were actively participated in this activity.
 - Through this activity, the students learn how to communicate with others, work as a team, and share their thoughts. As a result, the lesson was conveyed to the students that working in a team will result in a better solution.
 - Slow learners are also able to perform hash table implementation.

- ❖ ***Challenges faced in implementation:***

- This activity takes time and difficulties to organize the students into groups.
 - Some students in the team did not participate actively in the team.

References:

- ❖ <https://www.jigsaw.org/>
 - ❖ <https://www.readingrockets.org/strategies/jigsaw>

Signature of Faculty Member

(Dr.V. Anusuya)

HOD

(Dr.K.Vijayalakshmi)